
Experiment 7

Plate Heat Exchanger



Chemical Engineering Laboratory Report

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1 Abstract

The performance of a plate heat exchanger was studied by changing the hot-oil set temperature over seven operating points while maintaining an approximately constant cold-water flow. For every trial, the hot-oil flow rate was determined from the timed fall of liquid in a calibrated section, and the inlet and outlet temperatures of both streams were recorded after steady state was approached. These readings were used to calculate the heat duty, log-mean temperature difference (LMTD), and overall heat transfer coefficient. The measured value of U varied from 520.14 to 978.58 kcal h⁻¹ m⁻² °C⁻¹. A log-log regression gave $\log_{10}(U) = 0.4988 \log_{10}(\dot{m}) + 1.8419$, with $R^2 = 0.2558$, indicating a positive but weak relationship over the limited experimental range.

2 Aim

To determine the overall heat transfer coefficient of a plate heat exchanger at different hot-fluid operating conditions and to examine its dependence on the hot-fluid mass flow rate using a plot of $\log_{10}(U)$ against $\log_{10}(\dot{m})$.

3 Apparatus and Experimental Setup

3.1 Apparatus

The experimental unit consisted of the following:

- a corrugated-plate heat exchanger with an effective heat-transfer area of 0.0306 m²;
- an insulated hot-oil tank fitted with an electric immersion heater and PID temperature controller;
- a circulation pump for the hot-fluid loop;
- a regulated cooling-water supply;
- RTD sensors for the inlet and outlet temperatures of both streams;
- a graduated glass measuring section for determining the hot-oil flow rate;
- valves, connecting lines, drainage arrangement, and a stopwatch.

3.2 Flow Arrangement

The hot oil was heated in the insulated reservoir and circulated through alternate channels of the plate pack. Cooling water passed through the neighbouring channels, so the two fluids exchanged heat through the thin metal plates without mixing. The corrugations increase the effective area and disturb the boundary layers, producing stronger convective heat transfer than would be obtained from smooth, straight channels.

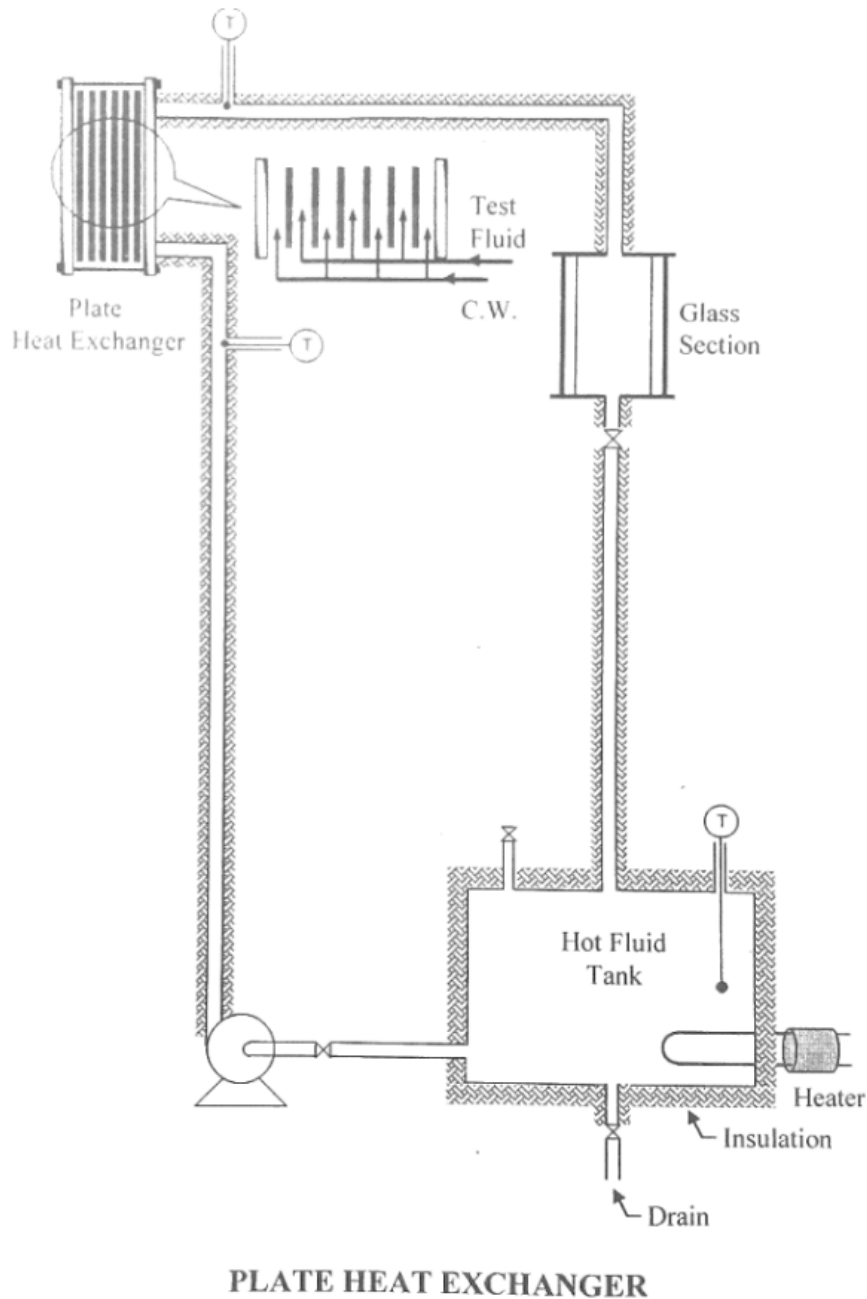


Figure 1: Schematic of the plate heat exchanger experimental unit.

4 Experimental Procedure

1. The main supply, hot-oil circulation pump, and temperature measurement system were switched on.
2. Cooling water was admitted to the exchanger and adjusted to approximately 430 L h^{-1} ; this setting was kept unchanged during the trials.
3. The required hot-fluid temperature was entered on the PID controller, beginning with a set point of 60°C .
4. After the displayed temperatures became nearly steady, the time required for the oil level in the measuring section to fall through 9 cm was measured.

5. The hot-fluid inlet and outlet temperatures and the cold-fluid inlet and outlet temperatures were recorded.
6. The set point was increased successively up to 90 °C, and the timing and temperature measurements were repeated, giving seven observations.
7. After completion, the heater was turned off, followed by the pump, water supply, and main electrical supply.

5 Theory

5.1 Heat Transfer in a Plate Heat Exchanger

A plate heat exchanger contains a series of thin, corrugated plates clamped together in a frame. Gaskets or welded joints direct the hot and cold fluids through alternate passages. The thin plate wall offers little conductive resistance, while the narrow, corrugated passages promote mixing and turbulence. Consequently, plate exchangers are compact and commonly achieve higher overall heat transfer coefficients than conventional shell-and-tube exchangers used for the same liquid–liquid service.

For a differential resistance network across one plate, the overall resistance may be written as

$$\frac{1}{U} = \frac{1}{h_h} + \frac{\delta_p}{k_p} + \frac{1}{h_c}, \quad (1)$$

where h_h and h_c are the hot- and cold-side film coefficients, δ_p is the plate thickness, and k_p is the thermal conductivity of the plate material.

5.2 Energy Balance

If heat loss to the surroundings is neglected, the heat lost by the hot oil is equal to the heat gained by the cooling water. The hot-side heat duty used in this report is

$$\dot{Q} = \dot{m}_h C_{p,h} (T_{h,i} - T_{h,o}). \quad (2)$$

5.3 Log-Mean Temperature Difference

Because the temperature difference between the two streams changes along the exchanger, an average arithmetic temperature difference is not appropriate. With the terminal temperature differences defined by

$$\Delta T_1 = T_{h,i} - T_{c,i}, \quad \Delta T_2 = T_{h,o} - T_{c,o}, \quad (3)$$

the log-mean temperature difference is

$$\Delta T_{\text{LMTD}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}. \quad (4)$$

The overall coefficient is then obtained from

$$\dot{Q} = U A \Delta T_{\text{LMTD}} \quad \Rightarrow \quad U = \frac{\dot{Q}}{A \Delta T_{\text{LMTD}}}. \quad (5)$$

5.4 Dependence on Hot-Fluid Flow Rate

Convective heat transfer is commonly represented through a Nusselt-number correlation,

$$Nu = \frac{hD_e}{k} = CRe^a Pr^b. \quad (6)$$

For fixed geometry and approximately constant properties, this can be reduced to $h_h = K\dot{m}^a$. If the wall and cold-side resistances are nearly unchanged, an approximate power-law dependence may also be written for the overall coefficient:

$$U = K'\dot{m}^a. \quad (7)$$

Taking common logarithms produces the linear form

$$\log_{10}(U) = \log_{10}(K') + a \log_{10}(\dot{m}). \quad (8)$$

Thus, the slope of a $\log_{10}(U)$ versus $\log_{10}(\dot{m})$ graph gives the empirical flow-rate exponent for the measured dataset.

6 Observations

Table 1: Fixed quantities used for data reduction.

Quantity	Value
Cross-sectional area of measuring tank, A_t	44 cm ²
Level drop in measuring section, L	9 cm
Heat-transfer area, A	0.0306 m ²
Specific heat of hot oil, $C_{p,h}$	0.648 kcal kg ⁻¹ °C ⁻¹
Density of hot oil, ρ_h	0.835 g cm ⁻³
Dynamic viscosity of hot oil (reference value)	3.46 cP
Cooling-water flow setting	approximately 430 L h ⁻¹

Table 2: Experimental readings recorded after approaching steady state.

Obs.	Set temp. (°C)	Time (s)	$T_{h,i}$ (°C)	$T_{h,o}$ (°C)	$T_{c,i}$ (°C)	$T_{c,o}$ (°C)	Level drop (cm)
1	60.0	9.86	57.1	51.1	23.7	25.2	9
2	65.0	7.78	62.9	54.8	23.8	25.5	9
3	70.0	7.36	67.5	56.9	23.8	25.7	9
4	75.0	8.75	73.0	60.4	23.8	25.9	9
5	80.5	11.71	77.5	62.7	23.9	25.9	9
6	85.7	14.50	80.2	63.7	24.0	26.3	9
7	90.0	10.50	84.2	64.8	24.1	26.5	9

7 Calculations

7.1 Flow-Rate Calculation

The fixed volume displaced during every timing measurement was

$$V = A_t L = (44)(9) = 396 \text{ cm}^3. \quad (9)$$

For observation 1,

$$\dot{V} = \frac{396}{9.86} \times \frac{3600}{10^6} = 0.144584 \text{ m}^3 \text{ h}^{-1}, \quad (10)$$

$$\dot{m}_h = \rho_h \dot{V} = (835)(0.144584) = 120.73 \text{ kg h}^{-1}. \quad (11)$$

7.2 Heat Duty, LMTD, and Overall Coefficient

Using observation 1 as a sample,

$$\dot{Q} = (120.7278)(0.648)(57.1 - 51.1) = 469.39 \text{ kcal h}^{-1}, \quad (12)$$

$$\Delta T_1 = 57.1 - 23.7 = 33.4 \text{ }^\circ\text{C}, \quad (13)$$

$$\Delta T_2 = 51.1 - 25.2 = 25.9 \text{ }^\circ\text{C}, \quad (14)$$

$$\Delta T_{\text{LMTD}} = \frac{33.4 - 25.9}{\ln(33.4/25.9)} = 29.49 \text{ }^\circ\text{C}, \quad (15)$$

$$U = \frac{469.39}{(0.0306)(29.49)} = 520.14 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^\circ\text{C}^{-1}. \quad (16)$$

The same procedure was applied to the remaining six observations.

Table 3: Calculated flow rate, heat duty, temperature differences, LMTD, and overall coefficient.

Obs.	$\dot{V}$ ($\text{m}^3 \text{ h}^{-1}$)	$\dot{m}_h$ (kg h^{-1})	$\dot{Q}$ (kcal h^{-1})	ΔT_1 ($^\circ\text{C}$)	ΔT_2 ($^\circ\text{C}$)	ΔT_{LMTD} ($^\circ\text{C}$)	U ($\text{kcal h}^{-1} \text{ m}^{-2} \text{ }^\circ\text{C}^{-1}$)
1	0.144584	120.73	469.39	33.4	25.9	29.49	520.14
2	0.183239	153.00	803.09	39.1	29.3	33.96	772.71
3	0.193696	161.74	1110.93	43.7	31.2	37.10	978.58
4	0.162926	136.04	1110.76	49.2	34.5	41.42	876.46
5	0.121742	101.65	974.91	53.6	36.8	44.67	713.15
6	0.098317	82.09	877.76	56.2	37.4	46.16	621.37
7	0.135771	113.37	1425.19	60.1	38.3	48.38	962.60

The quantities in Table 3 are rounded only for presentation. All calculations, logarithms, and the linear regression were performed using the unrounded spreadsheet values shown in Figure 2.

For reference, multiplying U in the above units by 1.163 converts it to $\text{W m}^{-2} \text{ K}^{-1}$; therefore, the measured range corresponds approximately to 605–1138 $\text{W m}^{-2} \text{ K}^{-1}$.

7.3 Spreadsheet Calculation Sheet

The complete spreadsheet record is included below to document the raw readings, unrounded calculated quantities, and the log-log regression used in this report.

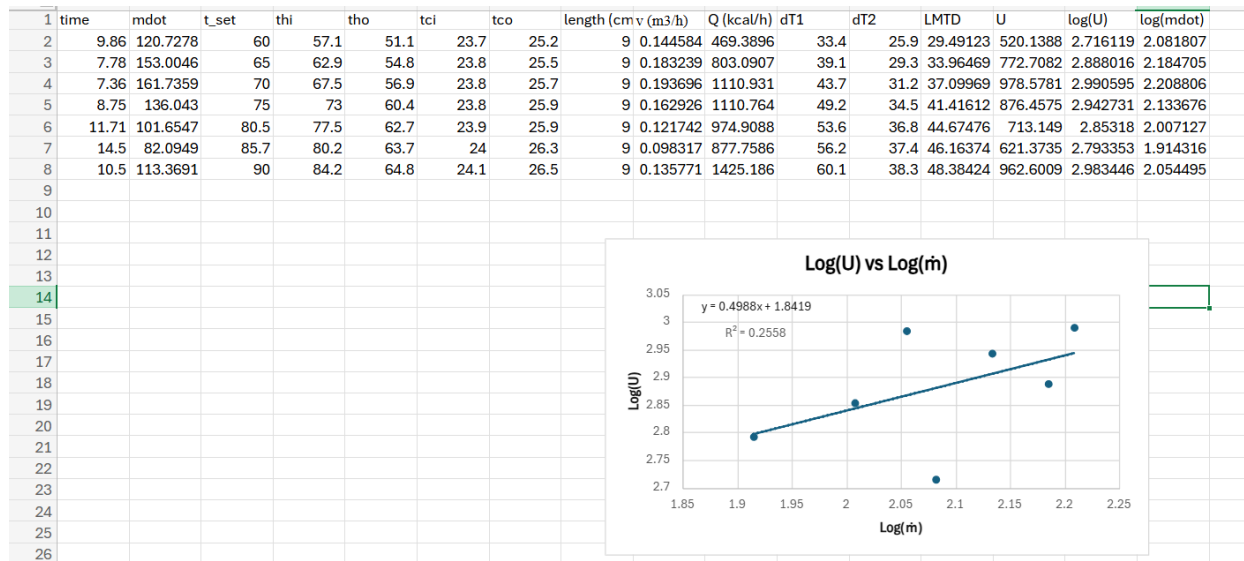


Figure 2: Excel calculation sheet containing the experimental readings, unrounded derived values, and the correctly oriented $\log_{10}(U)$ versus $\log_{10}(\dot{m}_h)$ regression.

8 Results and Discussion

Table 4: Logarithmic values used for regression.

Obs.	$\log_{10}(\dot{m}_h)$	$\log_{10}(U)$
1	2.0818	2.7161
2	2.1847	2.8880
3	2.2088	2.9906
4	2.1337	2.9427
5	2.0071	2.8532
6	1.9143	2.7934
7	2.0545	2.9834

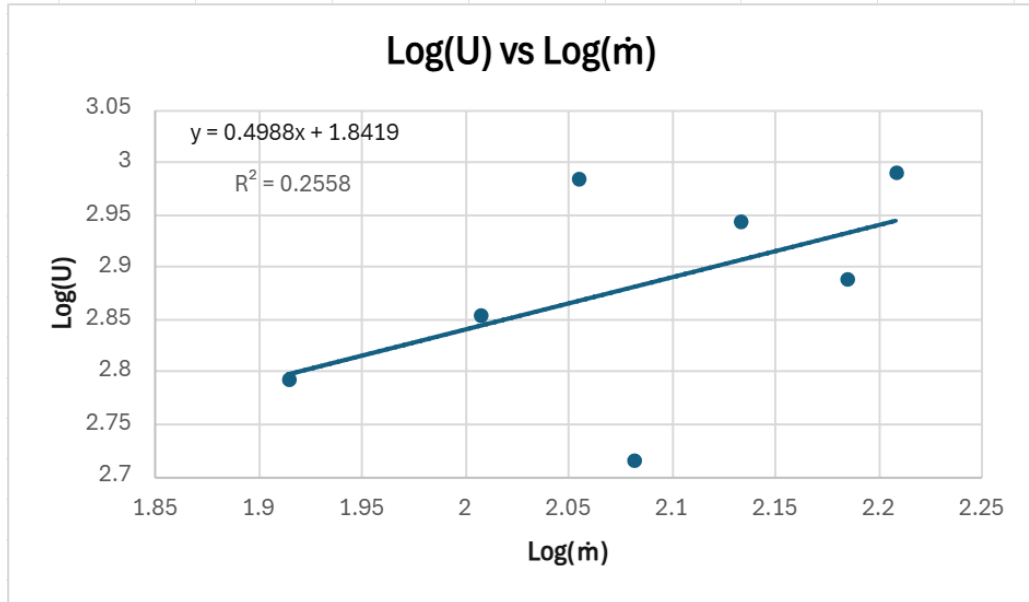


Figure 3: $\log_{10}(U)$ versus $\log_{10}(\dot{m}_h)$ with the linear trendline.

The least-squares fit obtained from the seven observations is

$$\log_{10}(U) = 0.4988 \log_{10}(\dot{m}_h) + 1.8419, \quad R^2 = 0.2558. \quad (17)$$

Equivalently,

$$U = 69.48 \dot{m}_h^{0.4988}, \quad (18)$$

for the units used in this experiment.

The positive slope indicates that an increase in hot-fluid mass flow rate generally improved the overall heat transfer coefficient. A higher flow rate reduces the fluid residence time but also increases the Reynolds number, thins the thermal boundary layer, and enhances the hot-side convection coefficient. The fitted exponent of approximately 0.50 is therefore qualitatively consistent with a flow-dependent convective process.

However, the low value of R^2 means that only about 25.6% of the variation in $\log_{10}(U)$ is described by this single linear relationship. The readings show clear scatter: for example, observations 4 and 7 have smaller flow rates than observation 3 but still give comparatively large U values because their temperature levels and heat duties are different. Hence, flow rate was not the only changing variable.

The oil temperature increased substantially from one run to the next, altering its viscosity and other properties, while small fluctuations in cold-water flow and temperature could also affect the cold-side film resistance.

The measured heat duty rose overall from 469.39 to 1425.19 kcal h⁻¹ as the hot-fluid set point increased. Meanwhile, U did not increase monotonically with set temperature: it reached its highest value in observation 3, decreased through observations 4–6, and increased again in observation 7. This behaviour reinforces that the experiment combined changes in flow, temperature-dependent properties, and measurement variability. A cleaner estimate of the exponent a would require the hot and cold inlet temperatures to be held constant while only the hot-fluid flow rate is varied deliberately.

9 Sources of Error

- **Manual timing:** Human reaction time in starting and stopping the stopwatch directly affects the calculated volumetric and mass flow rates.
- **Approach to steady state:** If a reading was taken while the inlet or outlet temperature was still drifting, the calculated LMTD and U would not represent a true steady operating point.
- **Heat loss:** Heat lost through the reservoir, pipework, fittings, and exchanger surface causes the hot-side heat duty to differ from the heat actually received by the water.
- **Property variation:** A single value of density and specific heat was used even though oil properties change with temperature. Viscosity, in particular, can vary strongly over the tested range.
- **Sensor error:** Calibration error, finite RTD resolution, and sensor lag influence both terminal temperature differences and therefore the LMTD.
- **Cold-flow variation:** The cooling-water setting was treated as constant, although minor supply-pressure fluctuations may have changed the cold-side coefficient between observations.

10 Assumptions

- The exchanger was at steady state when each set of four temperatures was recorded.
- Heat loss to the surroundings was negligible for the energy-balance calculation.
- The hot oil had uniform and constant density and specific heat at all operating points.
- The effective heat-transfer area remained 0.0306 m² and was fully available for heat exchange.
- Fouling resistance and the conductive resistance of the plate wall remained constant during the experiment.
- There was no leakage or mixing between the hot and cold streams.
- The cooling-water flow rate was constant at its set value.

11 Applications

Plate heat exchangers are preferred where compact equipment and efficient liquid–liquid heat transfer are required. Typical applications include:

- pasteurisation, heating, and cooling in dairy, beverage, and food-processing plants;
- heat recovery and utility integration in chemical and pharmaceutical plants;
- chilled-water loops, district heating, and HVAC installations;
- condensers and evaporators in refrigeration and heat-pump systems;
- cooling of lubricating oil, hydraulic oil, and engine coolant;
- marine duties and closed-loop cooling systems where space is limited.

12 Conclusion

The overall heat transfer coefficient of the plate heat exchanger was evaluated from seven sets of hot- and cold-stream temperature readings. The hot-oil mass flow rate ranged from 82.09 to 161.74 kg h⁻¹, while U ranged from 520.14 to 978.58 kcal h⁻¹ m⁻² °C⁻¹. Regression of the logarithmic data produced $\log_{10}(U) = 0.4988 \log_{10}(\dot{m}_h) + 1.8419$, corresponding to an empirical exponent of 0.4988. Although the positive slope agrees with the expected improvement of convective heat transfer at higher flow rates, the low R^2 of 0.2558 shows that the relationship was weak for this dataset. Simultaneous changes in temperature-dependent oil properties, possible cold-flow fluctuations, heat loss, and timing uncertainty likely contributed to the scatter. More controlled trials at fixed inlet temperatures and independently adjusted hot-fluid flow rates would provide a more reliable correlation.

13 References

1. CL 326: Integrated Chemical Engineering Laboratory Manual, “Plate Heat Exchanger” experiment, IIT Gandhinagar.
2. Experimental readings and calculations recorded by the group during the laboratory session.

14 Acknowledgement

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